

Shenzhen Dudumiao Technology Co., Ltd.

# smokefire Fire, Smoke & Smoking Detection

## Model Capability Test Guide

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Applicable software version: 1.0.0

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**Important**

This guide uses real images from public datasets to show actual model output and practical boundaries. smokefire is a video-AI assistance tool, not a certified fire alarm and not a substitute for statutory fire protection, patrols, emergency procedures, or human verification.

## Document information

| Item              | Value                                                                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Models            | <a href="#">fire_smoke_v5.pt</a> , <a href="#">smoking_v4.pt</a>                                                                                         |
| Default threshold | 0.30 for every image in this guide                                                                                                                       |
| Input size        | 640                                                                                                                                                      |
| Test sample       | 18 D-Fire images plus 12 CigDet images; 30 total                                                                                                         |
| Runtime           | NVIDIA CUDA validation environment with PyTorch 2.9.1+cu128 and Ultralytics 8.4.24; the validation-host model does not define customer GPU compatibility |
| Project           | <a href="https://github.com/newtv-ai/smokefire">https://github.com/newtv-ai/smokefire</a>                                                                |

## 1. Summary

| Fixed sample group      | Images | Actual result at 0.30                                                                 | Interpretation                                                          |
|-------------------------|--------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| D-Fire fire-only        | 5      | Fire boxes on 5 images                                                                | Responded to visible flames and several distant/night fire points       |
| D-Fire smoke-only       | 5      | Smoke boxes on 2; no detection on 3                                                   | Distant, thin, or low-contrast smoke remains a clear boundary           |
| D-Fire fire-and-smoke   | 4      | At least one class on all 4; both classes on 3                                        | Fire and smoke are evaluated separately in one scene                    |
| D-Fire neither          | 4      | No box on 3; smoke <a href="#">0.49</a> on 1                                          | Clouds, haze, glare, and similar patterns can cause false alerts        |
| CigDet positive smoking | 12     | Smoking boxes on all 12; top scores from <a href="#">0.35</a> to <a href="#">0.94</a> | Stronger on close views with visible hand/cigarette/mouth relationships |

These counts describe only the 30 fixed images shown here. They are not overall accuracy, recall, false-alert rate, or a project guarantee. Public datasets can differ materially from the target cameras.

## 2. Data sources and method

### 2.1 Public sources

| Dataset | Use                                       | Official location                                                                                                                         | License   |
|---------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| D-Fire  | Fire, smoke, combined, and neither images | <a href="https://github.com/gaia-solutions-on-demand/DFireDataset">https://github.com/gaia-solutions-on-demand/DFireDataset</a>           | CC0 1.0   |
| CigDet  | Real smoking/cigarette scenes             | <a href="https://data.mendeley.com/datasets/6hyrr8typ7/1">https://data.mendeley.com/datasets/6hyrr8typ7/1</a> , DOI 10.17632/6hyrr8typ7.1 | CC BY 4.0 |

D-Fire images were read from samples tagged `dfire` in the public mirror at <https://huggingface.co/datasets/Hajorda/flameye-wildfire-detection>; the official source and license remain the D-Fire repository above. CigDet is credited to Ali Khan, and this guide uses images from its public test set.

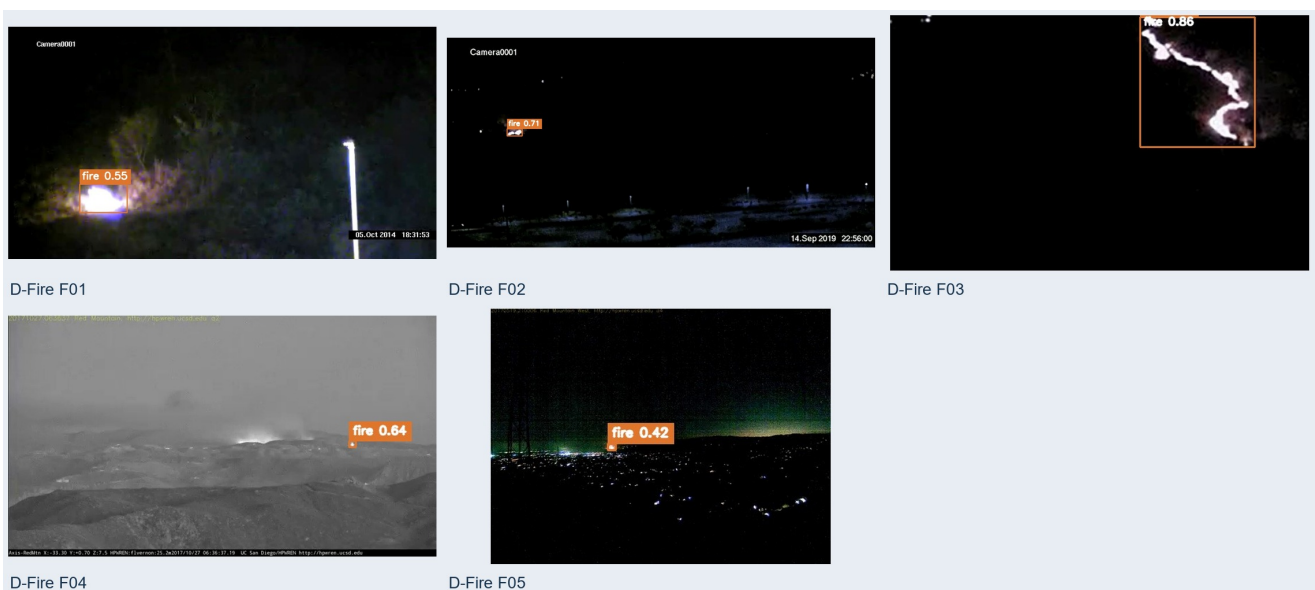
## 2.2 Fixed selection and inference

- D-Fire images were sorted by public image ID, then evenly sampled as 5 / 5 / 4 / 4 from fire-only, smoke-only, combined, and neither groups.
- CigDet test images were sorted by filename, then 12 images were selected at even intervals from the 111-image test set.
- Both delivered models used input size 640 and the default 0.30 threshold. Only model-returned boxes at or above 0.30 were drawn.
- No box was selected or added by hand. Misses remain unboxed and false positives remain visible.

### Important

Images were downloaded from real public datasets and may differ from the target site. Every box was generated by actual inference with `fire_smoke_v5.pt` or `smoking_v4.pt`; no box was added manually, and confidence is displayed to two decimal places.

## 3. Fire results



D-Fire fixed fire-only samples F01-F05

| ID  | Public image ID          | Highest actual output |
|-----|--------------------------|-----------------------|
| F01 | dfire_AoF00001           | fire 0.55             |
| F02 | dfire_AoF04540           | fire 0.71             |
| F03 | dfire_AoF04591           | fire 0.86             |
| F04 | dfire_PublicDataset00007 | fire 0.64             |
| F05 | dfire_PublicDataset00358 | fire 0.42             |

All five produced a box. F05 reached only 0.42, showing that confidence and stability can fall for distant small targets.

## 4. Smoke results

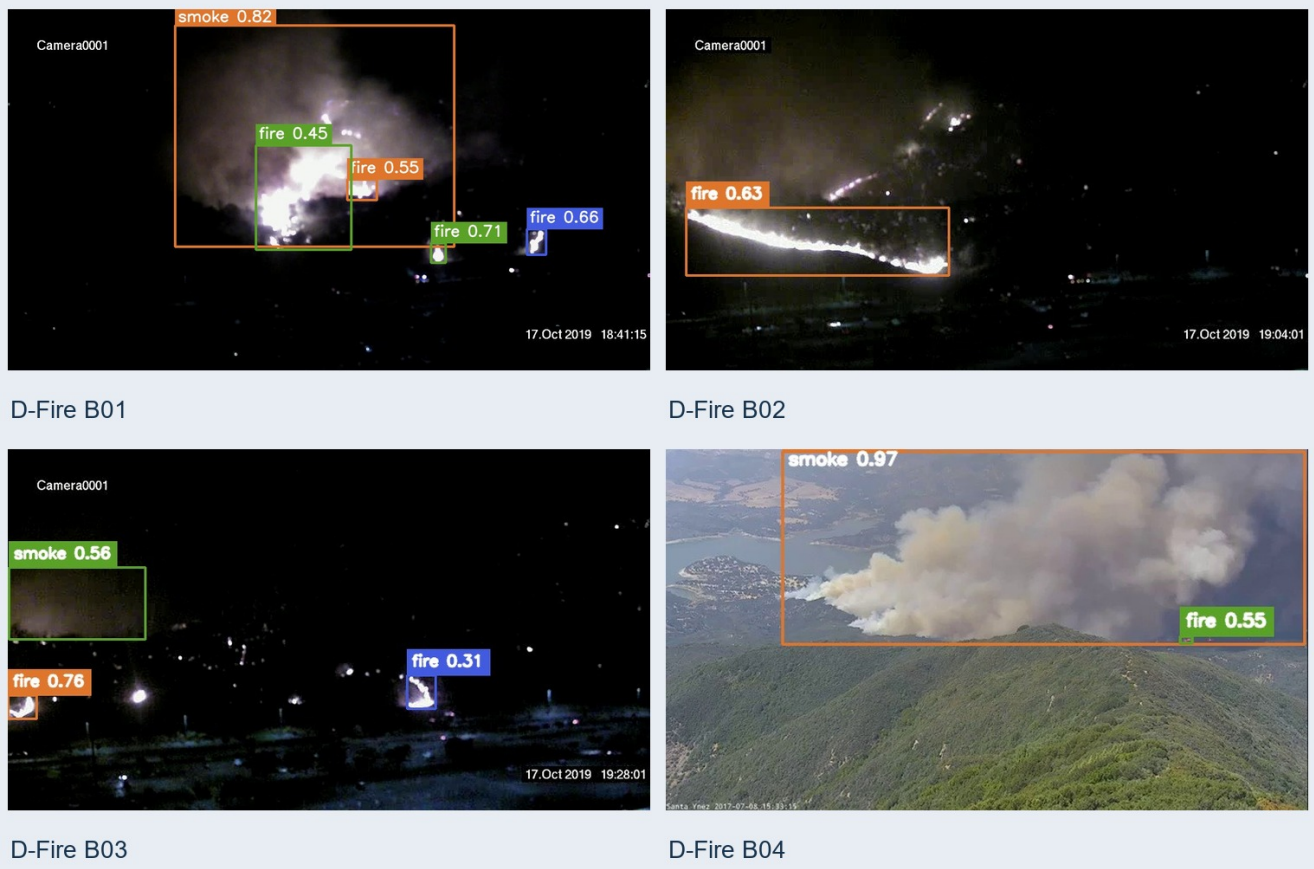


D-Fire fixed smoke-only samples S01-S05

| ID  | Public image ID          | Highest actual output |
|-----|--------------------------|-----------------------|
| S01 | dfire_AoF00002           | No detection          |
| S02 | dfire_AoF06061           | No detection          |
| S03 | dfire_PublicDataset00162 | No detection          |
| S04 | dfire_PublicDataset00469 | smoke 0.81            |
| S05 | dfire_PublicDataset00648 | smoke 0.93            |

Large plumes with clear boundaries and background contrast scored strongly. Distant, thin, low-contrast smoke or smoke that resembles cloud/haze can be missed.

## 5. Combined fire/smoke results



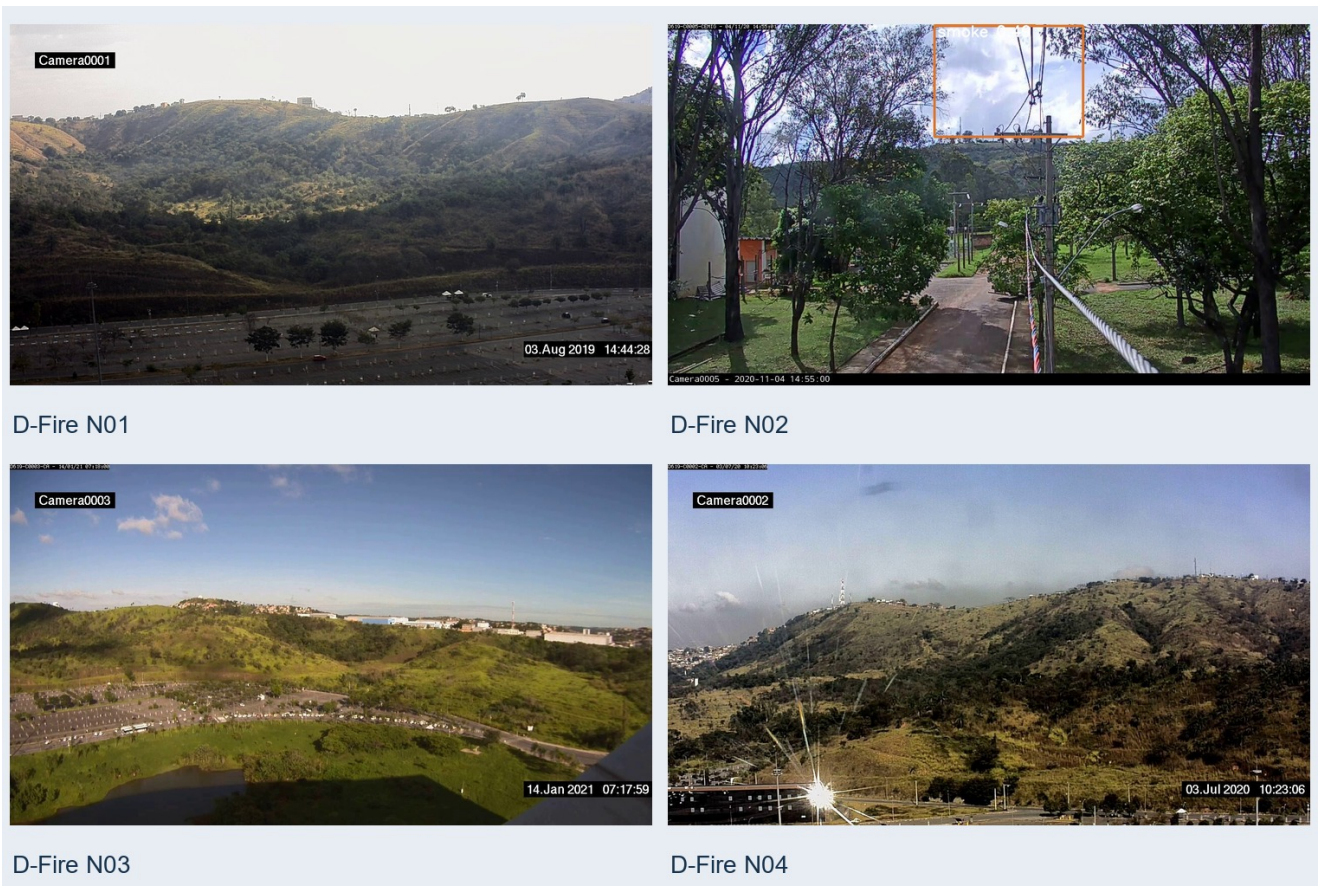
D-Fire fixed combined samples B01-B04

| ID  | Public image ID          | Main actual outputs           |
|-----|--------------------------|-------------------------------|
| B01 | dfire_AoF04019           | smoke 0.82; fire up to 0.71   |
| B02 | dfire_AoF04467           | fire 0.63; smoke not detected |
| B03 | dfire_AoF07841           | fire 0.76; smoke 0.56         |
| B04 | dfire_PublicDataset00656 | smoke 0.97; fire 0.55         |

All four produced at least one class, and three produced both. B02 shows that a combined scene does not guarantee that both classes pass the threshold.



## 6. No-fire/no-smoke interference



D-Fire N01

D-Fire N02

D-Fire N03

D-Fire N04

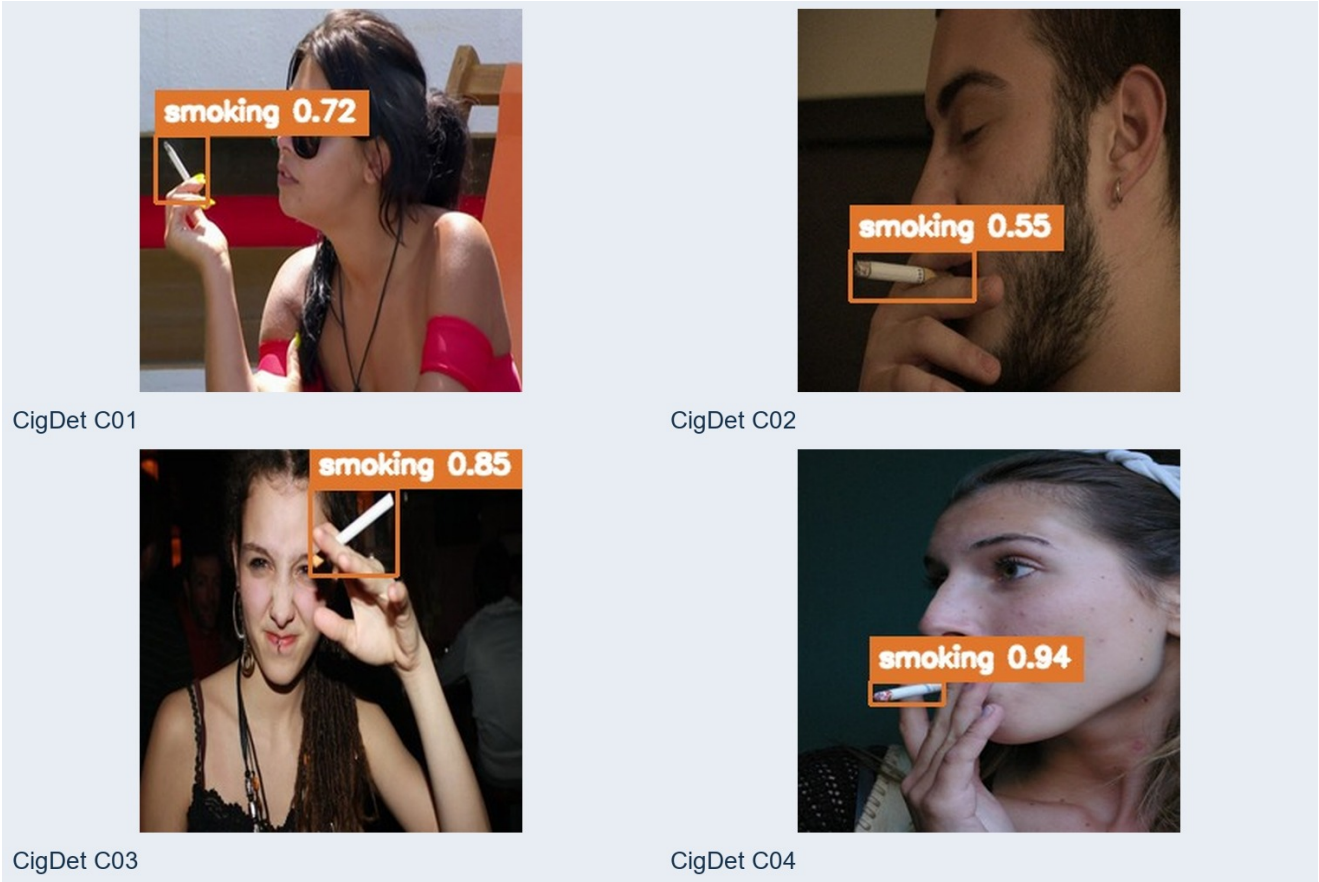
D-Fire fixed neither samples N01-N04

| ID  | Public image ID | Highest actual output |
|-----|-----------------|-----------------------|
| N01 | dfire_AoF00009  | No box                |
| N02 | dfire_AoF01530  | smoke 0.49            |
| N03 | dfire_AoF03575  | No box                |
| N04 | dfire_AoF07717  | No box                |

N02 is a retained real false-positive example. Site tests should include clouds, fog, steam, dust, reflections, welding, and night lighting rather than positive images alone.

## 7. Smoking results

### 7.1 C01-C04

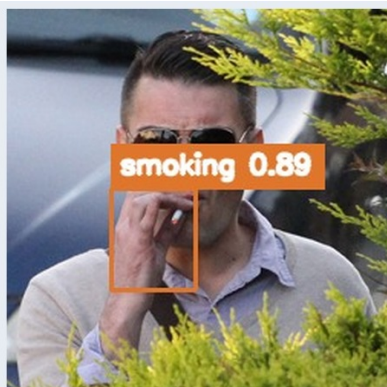


CigDet fixed samples C01-C04

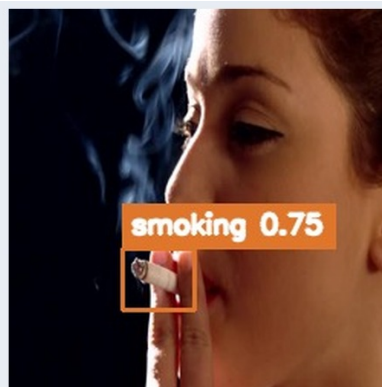
| ID  | Source file      | Highest actual output |
|-----|------------------|-----------------------|
| C01 | smoking_0020.jpg | smoking 0.72          |
| C02 | smoking_0068.jpg | smoking 0.55          |
| C03 | smoking_0122.jpg | smoking 0.85          |
| C04 | smoking_0196.jpg | smoking 0.94          |



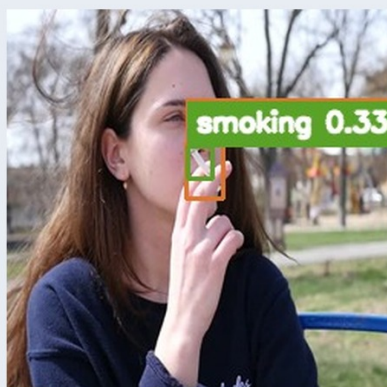
## 7.2 C05-C08



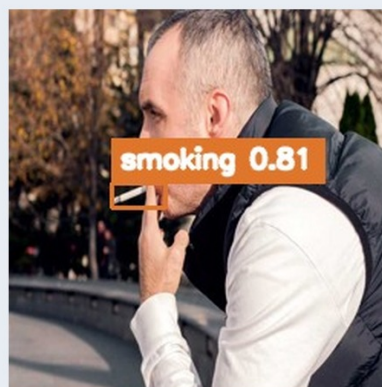
CigDet C05



CigDet C06



CigDet C07



CigDet C08

CigDet fixed samples C05-C08

| ID  | Source file      | Highest actual output |
|-----|------------------|-----------------------|
| C05 | smoking_0254.jpg | smoking 0.89          |
| C06 | smoking_0330.jpg | smoking 0.75          |
| C07 | smoking_0374.jpg | smoking 0.37          |
| C08 | smoking_0413.jpg | smoking 0.81          |

7.3 C09-C12



CigDet fixed samples C09-C12

| ID  | Source file      | Highest actual output |
|-----|------------------|-----------------------|
| C09 | smoking_0451.jpg | smoking 0.35          |
| C10 | smoking_0484.jpg | smoking 0.78          |
| C11 | smoking_0528.jpg | smoking 0.57          |
| C12 | smoking_0559.jpg | smoking 0.40          |

Every positive image produced a box at or above 0.30, but C07, C09, and C12 topped out at only 0.37 / 0.35 / 0.40. Smoking detection is sensitive to cigarette size, hand occlusion, pose, and distance. These close-view results must not be extrapolated directly to distant surveillance views.

8. Capability boundaries

- The model can analyze only what the camera captures. Blind spots, occlusion, overexposure, darkness, and compression loss cannot be fully recovered.
- Distant flames, cigarettes, and thin smoke may contain too few pixels even in a nominally high-resolution stream.

- Smoke is especially sensitive to clouds, fog, steam, dust, and background brightness; this fixed sample contains both misses and a false positive.
- CigDet is dominated by close views, while production CCTV is often high-angle and distant. Distribution shift directly changes results.
- Confidence is a relative model score for a candidate, not the probability that the real-world event is true, and scores are not comparable across models.
- A still image validates only one frame. Video alerts also depend on sampling interval, consecutive-frame policy, stream continuity, and compute load.

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## 9. Site acceptance

- 1 Build positive, ordinary-negative, and hard-negative samples for every representative camera view.
- 2 Cover day, night, backlight, rain/fog, distance, bitrate, and camera-angle variation.
- 3 Establish a baseline at the default 0.30 threshold. If it changes, replay the same set and measure both misses and false alerts.
- 4 Score whether a video event forms and record time-to-first-alert and evidence completeness.
- 5 Preserve software version 1.0.0, both model hashes, sample list, and results for reproducibility.
- 6 Keep human verification during pilot operation and expand only after agreed acceptance criteria are met.

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**<https://github.com/newtv-ai/smokefire>**